

Fidelity

Definition of Fidelity

→ Shows how much two quantum states overlap

$$F(\rho, \sigma) = \text{Tr}(\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}) \rightarrow \text{Root F: fidelity}$$

$$F(\rho, \sigma) = (\text{Tr}(\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}))^2$$

$$F(\rho, \sigma) = F(\sigma, \rho)$$

→ Fidelity in terms of trace norm

$$\|M\|_1 = \text{Tr}(\sqrt{M \cdot M^t}) = \text{Tr}(\sqrt{M^t \cdot M})$$

$$F(\rho, \sigma) = \|\sqrt{\rho} \sqrt{\sigma}\|_1 = \|\sqrt{\sigma} \sqrt{\rho}\|_1$$

$$\|M\|_1 = \max_{u \text{ is unitary}} |\text{Tr}(Mu)|$$

$$F(\rho, \sigma) = \max_{u \text{ is unitary}} |\text{Tr}(\sqrt{\sigma} \cdot \sqrt{\rho} \cdot u)|$$

→ Fidelity for Pure States

$$F(|\phi\rangle\langle\phi|, \sigma) = \sqrt{\langle\phi|\sigma|\phi\rangle}$$

$$F(|\phi\rangle\langle\phi|, |\psi\rangle\langle\psi|) = |\langle\phi|\psi\rangle|$$

$$F(\rho, \sigma) \in [0, 1]$$

Entanglement Fidelity

$$F_e(\rho) = \langle\phi^+|\rho|\phi^+\rangle$$

General Measurement Lemma

$$X \rightarrow \rho \rightarrow \text{Measured} \rightarrow \{P_0, \dots, P_{n-1}\}$$

$$\text{Say, } \text{Tr}(\rho_a \rho) > 1 - \varepsilon \quad \varepsilon \ll 1$$

$$F\left(\rho, \frac{\sqrt{\rho_a} \rho \sqrt{\rho_a}}{\text{Tr}(\rho_a \rho)}\right)^2 > 1 - \varepsilon \rightarrow \text{General Measurement Lemma}$$

Uhlmann's Theorem

$\rho, \sigma \rightarrow$ Represent states of X

Y has at least as many classical states as X

$$F(\rho, \sigma) = \max \{ \langle \phi | \psi \rangle : \text{Tr}_Y(|\phi\rangle\langle\phi|) = \rho, \text{Tr}_Y(|\psi\rangle\langle\psi|) = \sigma \}$$

Refer Handwritten
notes on GitHub
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